A structural obstruction to $(1, 1, 3, 3, 4)$-packing colorings

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Abstract

Mortada, El Zein and Al Hajjar asked whether every connected claw-free subcubic graph, apart from their specified exception, admits a $(1, 1, 3, 3, 4)$-packing coloring. We give a negative answer. For every odd integer $k \geq 3$, the triangle expansion of a ring of k diamonds is a connected cubic graph on $12k$ vertices that admits no such coloring. The proof selects an arbitrary existing high-colored vertex in each triangle. The selected labels give a proper three-coloring of the core, and the radius-four class forces an external private neighbor at each of its core vertices. This requirement excludes that class from every diamond tip, leaving an impossible two-coloring of an odd cycle. The proof is structural and requires no exhaustive search. A 36-vertex member is also certified in Lean using actual graph distances; that formalization does not cover the entire infinite family.

Keywords. Packing coloring; claw-free graph; cubic graph; triangle expansion; counterexample; formal verification.
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1 Introduction

A packing coloring assigns different distance requirements to its color classes. Repeated numbers in a prescribed sequence stand for distinct labels, each with its own class. This distinction is particularly useful for subcubic graphs, where several independent sets and a few more widely separated classes can be combined. For background on this framework, see Gastineau and Togni [4] and the survey of Brešar, Ferme, Klavžar and Rall [3].

Mortada, El Zein and Al Hajjar [5] proved that a connected claw-free subcubic graph other than a specified graph $\mathcal{H}$ admits a $(1, 1, 3, 3, 3)$-packing coloring. They then asked in Section 6, Problem 1 whether the third radius-three class could always be replaced by a radius-four class. Their exception $\mathcal{H}$ has 12 vertices. Our constructions have at least 36 vertices and satisfy all the other hypotheses of that question.

The answer is negative even for cubic graphs obtained by the standard operation of replacing each vertex of a cubic core by a triangle. Our cores are odd rings of diamonds. Neither these cores nor the triangle replacement operation is new: both occur in the structural description of claw-free cubic graphs, for example in Oum [7] and Brešar, Kuenzel and Rall [2]. The

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mathematical content here is the obstruction to the particular distance sequence $(1, 1, 3, 3, 4)$ and its proof for all odd ring lengths.

The key reduction uses only necessary conditions. Each replacement triangle must contain a vertex with one of the three high labels. Choosing any such vertex gives a proper three-coloring of the core. For the label of radius four, the chosen incidence must point to a core neighbor that is adjacent to no other core vertex with that label. An odd diamond ring has no proper three-coloring satisfying this additional requirement. No assumption about a unique high vertex per triangle, or a normal form for the original coloring, is needed.

We also give an exact, machine-checkable 36-vertex instance. Its Lean theorem uses Mathlib's `SimpleGraph`, walks and extended graph distance, and quantifies over every five-label coloring of the graph. The infinite-family proof in this paper is an ordinary mathematical proof. Computations provide reproducibility and controls, rather than a premise of the family theorem.

1.1 Related work and contribution boundaries

Brešar, Kuenzel and Rall [2] established $(1, 1, 2, 2)$-packing colorability for claw-free cubic graphs. Mortada and El Zein [6] subsequently gave a shorter proof. Anderson, Kuenzel and Marcano Cuellar [1] proved $(1, 1, 2, 3)$-packing colorability for diamond-free claw-free cubic graphs. The expanded graphs used here are themselves diamond-free: the diamonds lie in their cores. These positive results concern different distance sequences and are consistent with our obstruction.

The result of Mortada, El Zein and Al Hajjar [5, Theorem 4] also implies that every graph in our family admits a $(1, 1, 3, 3, 3)$-packing coloring. We use that known conclusion only to place the obstruction in context. We do not claim to refute that theorem, or to settle their separate $(1, 1, 2, 3)$ problem for general subcubic graphs. We make no claim of minimum counterexample order, an exact ordinary packing chromatic number, or historical firstness. The original arXiv record and related literature were checked in several bounded searches; absence of an earlier verified answer in those searches is not an exhaustive priority certification.

2 Definitions and main result

All graphs in this paper are finite, simple and undirected. For a graph F, its open neighborhood at v is $N_F(v)$. The distance $\mathrm{dist}_F(x, y)$ is the minimum length of an x–y path, with distance ∞ between vertices in different components. Distances in an induced subgraph are not used for packing constraints. A graph is subcubic if every vertex has degree at most three, cubic if every vertex has degree three, and claw-free if it contains no induced copy of $K_{1,3}$.

Definition 2.1. For a sequence $S = (s_1, \ldots, s_t)$ of positive integers, an S-packing coloring of F is a map $c : V(F) \longrightarrow \{1, \ldots, t\}$ such that

$$x \neq y, \quad c(x) = c(y) = j \quad \Longrightarrow \quad \mathrm{dist}_F(x, y) > s_j.$$

Equal entries of S correspond to different color labels. For $S = (1, 1, 3, 3, 4)$, we denote the labels by $1_a, 1_b, A, B, C$, with radii $1, 1, 3, 3, 4$, respectively. The labels A, B, C are called high labels.

Definition 2.2 (Triangle expansion)**.** Let D be a simple cubic graph. The graph $\mathcal{T}(D)$ has vertex set

$$V(\mathcal{T}(D)) = \{(v, w) : v \in V(D),\ w \in N_D(v)\}.$$

For each v, the three vertices $T_v = \{(v, w) : w \in N_D(v)\}$ form a triangle. In addition, for each edge $vw \in E(D)$, add the edge $(v, w)(w, v)$. There are no other edges. The pair (v, w) is an incidence, or port, at v pointing to w.

For an integer $k \geq 3$, take indices in $\mathbb{Z}/k\mathbb{Z}$. Define D_k on the $4k$ distinct vertices

$$a_i, b_i, p_i, q_i \qquad (i \in \mathbb{Z}/k\mathbb{Z})$$

with edge set

$$E(D_k) = \bigcup_i \{a_i b_i, a_i p_i, a_i q_i, b_i p_i, b_i q_i, q_i p_{i+1}\}. \tag{1}$$

Thus $\{a_i, b_i, p_i, q_i\}$ induces a diamond $K_4 - p_i q_i$, and the additional edges join the diamonds in a ring. Put $G_k = \mathcal{T}(D_k)$.

Theorem 2.3. *For every odd integer $k \geq 3$, the graph G_k is connected, cubic, claw-free and diamond-free, has $12k$ vertices and $18k$ edges, and admits no $(1, 1, 3, 3, 4)$-packing coloring.*

Corollary 2.4. *Problem 1 in Section 6 of Mortada, El Zein and Al Hajjar [5] has a negative answer. In particular, G_3 is a 36-vertex counterexample, and counterexamples exist with arbitrarily large order.*

Proof. Each G_k is a connected claw-free subcubic graph. Since $|V(\mathcal{H})| = 12$ and $|V(G_k)| = 12k \geq 36$, the graph G_k is not isomorphic to the exception $\mathcal{H}$. Odd integers $k \geq 3$ are unbounded. □

3 Structure and metric of the expansion

Lemma 3.1. *For every $k \geq 3$, D_k is a connected simple cubic graph. Its triangle expansion G_k is connected, cubic, claw-free and diamond-free, with $12k$ vertices and $18k$ edges.*

Proof. Within a diamond, a_i and b_i have neighbors equal to the other base vertex and both tips. The tips have neighborhoods

$$N_{D_k}(p_i) = \{a_i, b_i, q_{i-1}\}, \qquad N_{D_k}(q_i) = \{a_i, b_i, p_{i+1}\}. \tag{2}$$

All these neighbors are distinct for $k \geq 3$. The edges in (1) have no loops or repetitions. Every diamond is connected, and the joining edges form a ring of the diamonds, so D_k is connected.

More generally, for any simple cubic D, a vertex (v, w) of $\mathcal{T}(D)$ has two neighbors in T_v and the third neighbor (w, v) outside T_v. These three neighbors are distinct, and the first two are adjacent. Hence the expansion is simple, cubic and claw-free. A path in D lifts to a walk between its replacement triangles, and every triangle is connected; therefore connectedness of D implies connectedness of $\mathcal{T}(D)$. There are $3|V(D_k)| = 12k$ incidences, and cubicity gives $|E(G_k)| = 3(12k)/2 = 18k$.

The edges between distinct replacement triangles form a matching. No such edge can belong to a triangle: a hypothetical third vertex would require one of its endpoints to have a second neighbor outside its own replacement triangle. Thus every triangle of the expansion is one of the T_v. A diamond contains two triangles sharing an edge, whereas the T_v are vertex-disjoint. Consequently G_k is diamond-free. This last observation concerns the expanded graph; the core D_k certainly contains diamonds. □

Lemma 3.2. *Let D be a simple cubic graph.*

1. *If $uv \in E(D)$, then every pair $x \in T_u$, $y \in T_v$ is joined by a walk of length at most three in $\mathcal{T}(D)$.*

2. *If $x = (v, w)$, $u \in N_D(w) \setminus \{v\}$ and $y \in T_u$, then x and y are joined by a walk of length at most four in $\mathcal{T}(D)$.*

Proof. For the first assertion, use the internal step from x to (u, v), then the edge $(u, v)(v, u)$, and finally the internal step from (v, u) to y. An internal step is omitted when its endpoints coincide. Each internal step has length at most one.

For the second assertion, use

$$(v, w),\ (w, v),\ (w, u),\ (u, w),\ y. \tag{3}$$

The middle internal edge exists because $u \neq v$. The final internal step has length zero or one. These are walks in the full expansion, so the corresponding graph distances have the same upper bounds. No claim that the displayed walks are shortest is required. □

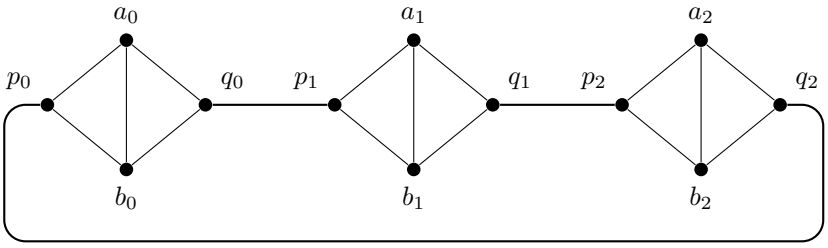


Figure 1: The core D_3. Each of its twelve vertices is replaced by a triangle to obtain G_3. The displayed diamonds are in the core, not in the 36-vertex expansion.

4 A necessary condition on the core

An external private neighbor of v with respect to a set I is a vertex $w \in N_D(v) \setminus I$ satisfying $N_D(w) \cap I = \{v\}$. We use this standard terminology only for the explicitly stated condition below; no domination theorem is invoked.

Proposition 4.1. *If $\mathcal{T}(D)$ admits a $(1, 1, 3, 3, 4)$-packing coloring, then D admits a proper three-coloring $f : V(D) \to \{A, B, C\}$ such that each vertex of $I = f^{-1}(C)$ has an external private neighbor with respect to I.*

Proof. Let c be any $(1, 1, 3, 3, 4)$-packing coloring of $\mathcal{T}(D)$. All labels have positive radius, so adjacent vertices have distinct labels. In particular, the triangle T_v cannot be colored solely with the two labels $1_a, 1_b$. Choose a vertex $x_v \in T_v$ having a high label, and put $f(v) = c(x_v)$. The choice can be arbitrary; T_v may contain more than one high-colored vertex.

If $uv \in E(D)$, the triangles T_u, T_v are disjoint, and Lemma 3.2 gives $\mathrm{dist}_{\mathcal{T}(D)}(x_u, x_v) \leq 3$. Each high label has radius at least three. Therefore $f(u) \neq f(v)$, and f is proper.

Take $v \in I$ and write its selected incidence as $x_v = (v, w)$. Then $w \in N_D(v)$, and properness shows $w \notin I$. Also $v \in N_D(w) \cap I$. If another vertex $u \in N_D(w) \cap I$, $u \neq v$, existed, then Lemma 3.2 and (3) would give $\mathrm{dist}_{\mathcal{T}(D)}(x_v, x_u) \leq 4$. But these are distinct vertices both assigned the label C, whose radius is four. The required strict inequality is impossible. Thus $N_D(w) \cap I = \{v\}$, as required. □

Remark 4.2. Proposition 4.1 is a necessary condition, not an equivalence or a coloring algorithm. It neither constructs the radius-three representatives nor ensures that the remaining vertices can be colored by the two low labels. The contradiction below needs only necessity. In particular, no restriction to colorings with exactly one high vertex per triangle is justified or used.

5 The obstruction in odd diamond rings

Lemma 5.1. *Let f be a proper three-coloring of D_k with labels A, B, C. For each i, the tips p_i, q_i have a common color t_i, and $t_i \neq t_{i+1}$. If every C-colored vertex has an external private neighbor with respect to $f^{-1}(C)$, then $t_i \neq C$ for every i.*

Proof. The adjacent bases a_i, b_i have different colors. Each tip is adjacent to both bases, so both tips have the third color, denoted by t_i. The joining edge $q_i p_{i+1}$ gives $t_i \neq t_{i+1}$.

Suppose that $t_i = C$, and put $I = f^{-1}(C)$. Both p_i and q_i belong to I. By (2), a possible private neighbor of p_i must be one of a_i, b_i, q_{i-1}. Neither a_i nor b_i is private, since each is also adjacent to $q_i \in I$, and $q_i \neq p_i$.

The remaining candidate is q_{i-1}. Its color is $t_{i-1} \neq t_i = C$, because $q_{i-1}p_i$ is an edge. In the preceding diamond, the two base colors and its tip color are all three labels. Consequently one of a_{i-1}, b_{i-1} has color C. This base vertex is adjacent to q_{i-1}, belongs to I, and is distinct from p_i. Hence q_{i-1} is not a private neighbor either. We have exhausted all three neighbors of p_i, a contradiction. □

Proof of Theorem 2.3. The graph-theoretic assertions follow from Lemma 3.1. Suppose G_k admitted a $(1, 1, 3, 3, 4)$-packing coloring. Proposition 4.1 would give a proper core three-coloring with the private-neighbor property. Lemma 5.1 would then give labels $t_i \in \{A, B\}$ with $t_i \neq t_{i+1}$ at every cyclic index. Starting with t_0, these labels alternate around the ring. Since k is odd, $t_{k-1} = t_0$, contradicting the last joining edge. Thus no such packing coloring exists. □

Remark 5.2 (Why radius four matters). The walk (3) has length at most four. For a radius-three label, this upper bound alone gives no contradiction: distance four is allowed. The private-neighbor conclusion for C therefore uses precisely the stronger distance requirement. By the known $(1, 1, 3, 3, 3)$-colorability result [5, Theorem 4], the members of our family do admit the weaker sequence. This is a comparison with that theorem, not a new proof of it.

6 The fixed certificate and formal verification

The full mathematical proof above is independent of computation. For a reproducible finite witness, we additionally specify G_3 as follows. Number the core vertices by

$$a_i = 4i, \qquad b_i = 4i + 1, \qquad p_i = 4i + 2, \qquad q_i = 4i + 3 \quad (0 \leq i < 3).$$

List the incidences (v, w) lexicographically, with neighbors of each v in increasing numerical order. Number them $0, \ldots, 35$. This convention determines the graph exactly, without relying on a picture or a stored distance table.

6.1 What the Lean theorem proves

The source `ClawFree11334.lean` implements this graph as a Mathlib `SimpleGraph (Fin 36)`. It proves agreement with the 54 undirected certificate edges, degree exactly three at every vertex, connectedness by actual walks, exclusion of every induced claw, and nonisomorphism to every graph on 12 vertices.

The packing predicate is a strict inequality between the radius $1, 1, 3, 3, 4$ of a label and the actual Mathlib extended graph distance. The main declaration is

`ClawFree11334.original_counterexample`.

It includes the failure of all color functions $\mathrm{Fin}(36) \to \mathrm{Fin}(5)$, not merely failure within a selected family of colorings. Intermediate declarations establish the actual length-three and length-four walks, extract arbitrary high representatives, and prove the core obstruction structurally. Small ordinary kernel decisions check fixed graph data and local three-color identities. The proof does not enumerate all five-color functions, use `native_decide`, admit a proof, or introduce a nonstandard axiom.

The supplied audit covers 18 declarations. Its allowed axioms are

`propext`, `Classical.choice`, `Quot.sound`.

The supplement binds the source, toolchain and actual compilation log by SHA256. The precise formalization boundary is important: the Lean theorem covers $k = 3$, and does not formalize all odd k, or the separate diamond-free observation in Lemma 3.1.

6.2 Reproducibility and independent checks

The supplement contains the fixed adjacency certificate, the exact Lean source, version information, and definition-first Python checks. Our bounded replay reconstructs the incidence graph from (1), compares it with the frozen certificate, computes full-graph BFS distances, and checks the short-walk bounds. It includes damaged-input and positive controls, as described in Appendix A.

An earlier author's exhaustive projected check is retained as historical evidence: it examined $3^{12} = 531441$ assignments, found 48 proper core labelings, and checked 3888 candidate selections for their radius-four representatives, finding none. These figures are explicitly author-provenanced and are not described as a new run during manuscript preparation. Neither Theorem 2.3 nor its Lean instance depends on that finite search.

An independent AI reviewer, uninvolved in discovery, implementation and manuscript writing, rederived the proof from the original definitions and checked the manuscript's mathematical and semantic claims. AI-assisted independent checking is not external human peer review. Version-bound review and execution records are included with the reproducibility supplement.

7 Discussion

The obstruction has two distinct components. Triangle expansion converts short walks into restrictions on a core coloring. The diamond ring then amplifies the local private-neighbor restriction into a global parity contradiction. This explains why the witness extends to arbitrarily large orders without a search or a new distance calculation for each graph.

The core condition is a useful way to test a proposed $(1, 1, 3, 3, 4)$-coloring route, but we have not proved that it is sufficient. Even ring lengths escape the final parity contradiction; that fact alone does not prove their expanded graphs colorable. An explicit $k = 4$ coloring is checked as a positive control in the supplement. We do not claim a classification of all admissible cores, the smallest counterexample, or a solution to other packing coloring problems.

The fixed formal witness is enough to give a negative answer to the original universal question. Extending the Lean development to cyclic indices of arbitrary odd order would formalize the family theorem, but is not needed for the mathematical proof presented here. The source's positive results for weaker palettes remain part of the existing literature and are not counted as additional new results of this paper.

AI assistance and responsibility

OpenAI Codex and a GPT-6-based agent system contributed substantially to mathematical exploration, construction and proof development, exact computation, Lean formalization, literature searches, independent AI checks, and manuscript writing. Separate AI reviewers examined the mathematical arguments and their correspondence to the original problem. Such review does not constitute external human peer review or certify historical originality. No AI system is listed as an author. River Zhang is the named human author; scientific responsibility for the final manuscript and any submission remains with the human author. No human-only research process is asserted.

A Exact replay and certificate conventions

The fixed JSON certificate uses the numbering given in Section 6. Undirected edge pairs are stored once, with the smaller endpoint first, in lexicographic order. The incidence labels and the palette are part of the certificate, not descriptive metadata that a checker may ignore. The following pseudocode reconstructs the object.

> For each cyclic index i, add the five diamond edges and $q_i p_{i+1}$. For each core vertex v, sort its three neighbors and list the incidences (v, w). Join all three pairs within T_v, and join each incidence to its reciprocal (w, v). Reject loops, repeated edges and a mismatching certificate. Compute BFS from every expanded vertex.

The bounded structural replay checks cubicity, connectivity, claw-freeness, unique replacement triangles, and all the metric bounds used in Proposition 4.1. It checks the tip-neighbor formulas at every cyclic index without enumerating the full space of original colorings. A separate independent checker reconstructs the graph without importing the author's verifier. The finite regressions are checks of the implementation, not proofs of the all-odd-k statement.

The controls include a missing edge, a corrupted incidence label, a changed palette, an invalid equal-low-label coloring on adjacent vertices, a valid $(1, 1, 3, 3, 3)$ coloring of G_3, and a valid $(1, 1, 3, 3, 4)$ coloring of G_4. Every positive coloring is tested against distances in the entire expanded graph. The historical exhaustive verifier remains available as an optional command; it is not part of the bounded publication replay.

For Lean, use the supplied `lean-toolchain` and `lake-manifest.json`; the evidence uses Lean 4.34.1 and Mathlib commit

`d13f23b723b8a846827a245b89c10fc7d3f11612`.

The `REPRODUCE.md` file gives the exact commands, including the standalone source-root option. The delivery manifest binds the paper, source, certificate, reviews and execution records by SHA256. Installing Lean from an empty cache is a separate environment step and is not claimed as a tested part of this package replay.

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